

Artificial Intelligence

Lesson 20: Bayesian Networks

Structure → Local Probabilities → Factorization → Independence

Graph
structure

CPTs
local probabilities

Factorization
compact joint

Independence
assumptions

A Bayesian network is a compact probability model: graph structure plus local probability tables.



From one uncertain variable to many

L18 and L19 handled single questions. Real AI systems connect many uncertain variables.

L18

$$P(X)$$

distribution over one uncertain
variable

L19

$$P(X | E)$$

updated belief after evidence

L20

many variables

connected uncertainty model

Today: represent how uncertain variables influence one another.

The full joint grows quickly

A complete joint table lists every possible assignment of every variable.

4 binary variables

$$2^4 = 16$$

joint assignments

10 binary variables

$$2^{10} = 1024$$

joint assignments

Problem

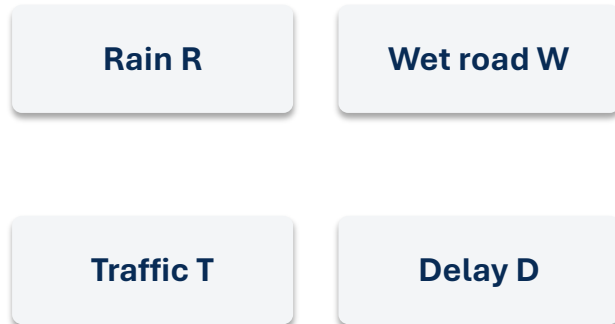
most combinations are not equally independent

A Bayesian network avoids writing one giant table when structure tells us what depends on what.

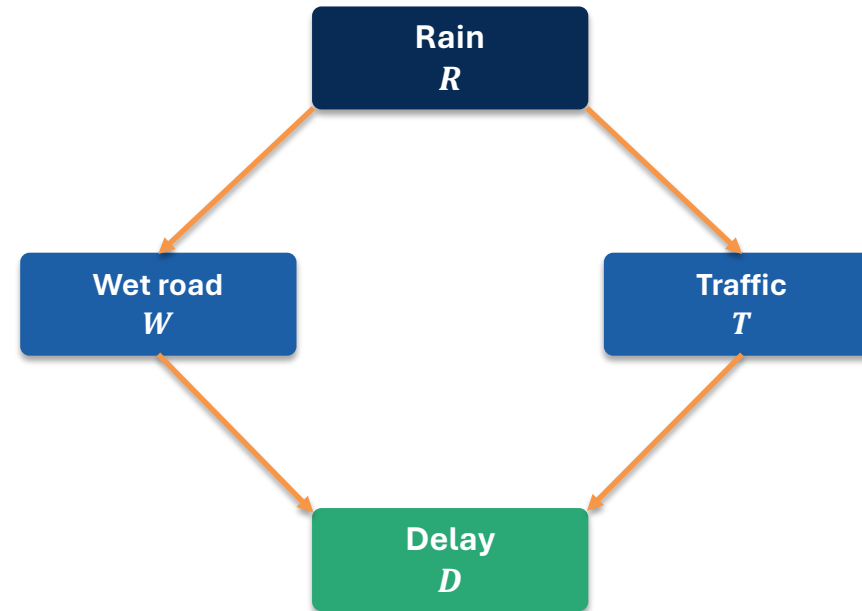
Structure can reveal what depends on what

The graph is a compact claim about direct probabilistic relationships.

Unstructured variables



Structured model



Question: Does Delay need to depend directly on Rain once Wet Road and Traffic are already known?

Today's mission

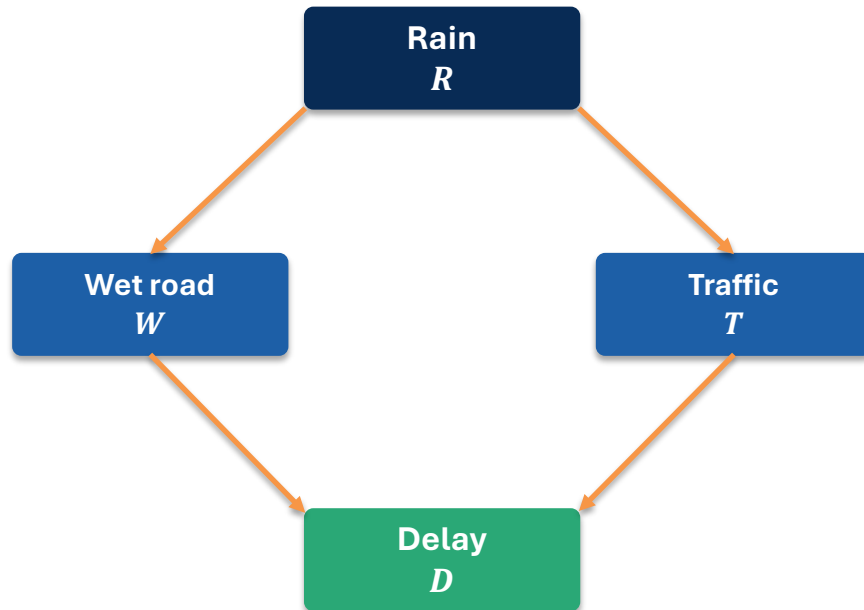
Bayesian networks connect graph structure with probability tables.

- 1 Construct and interpret a Bayesian network from a small scenario.
- 2 Identify parents, children, ancestors, descendants, and simple conditional-independence relationships.
- 3 Associate each node with the appropriate prior or conditional probability table.
- 4 Factor the full joint distribution according to the network structure.

Working rule: scenario $\leftrightarrow$ graph $\leftrightarrow$ CPTs $\leftrightarrow$ factorization $\leftrightarrow$ independence statement

Primer: a Bayesian network is a DAG

The graph vocabulary matters before the probabilities appear.



**Directed
arrows have
direction**

**Acyclic
following arrows cannot
loop back**

**Graph
nodes connected by
edges**

DAG = directed acyclic graph. It is the skeleton of the probability model.

Nodes are random variables

A node represents an uncertain variable, not one specific value.

Rain
R

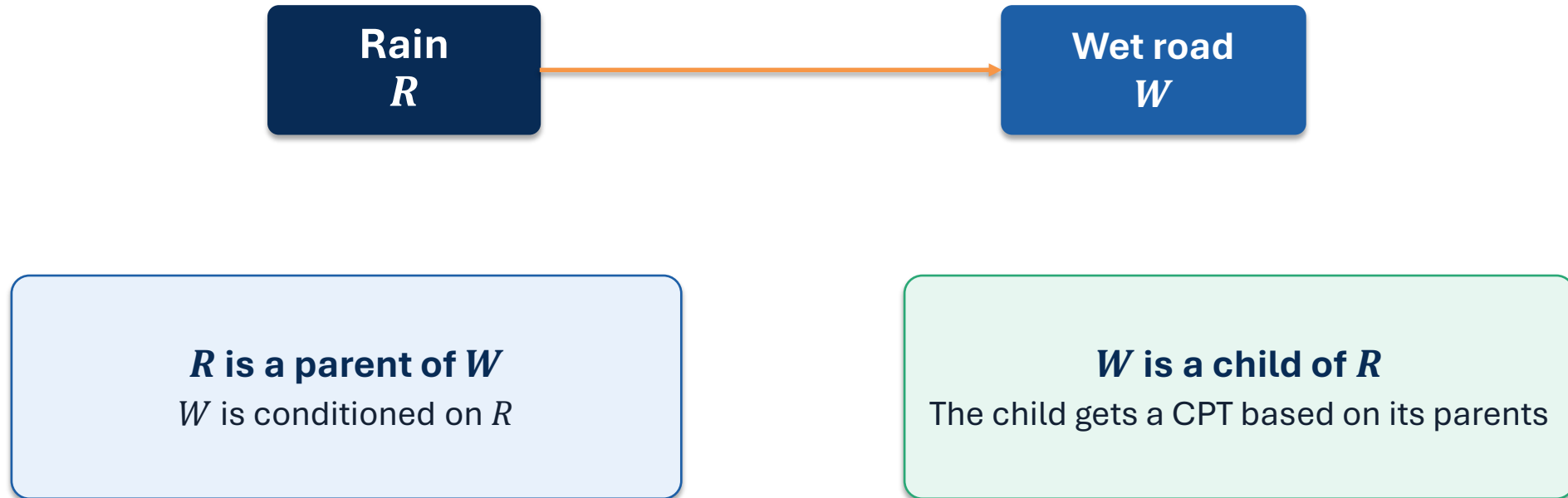
Random variable
R = Rain
Can be true or false

Assignment
R = true
One specific value of the
variable

This distinction matters when we later write full assignments such as $P(r, w, t, d)$.

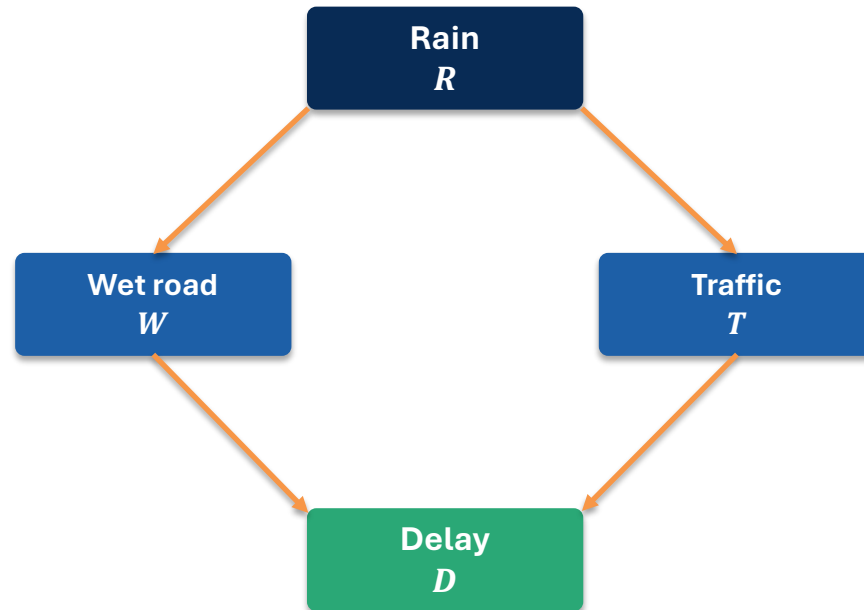
Parents and children

An arrow creates a local relationship between two variables.



Ancestors and descendants

Longer paths show indirect relationships in the network.



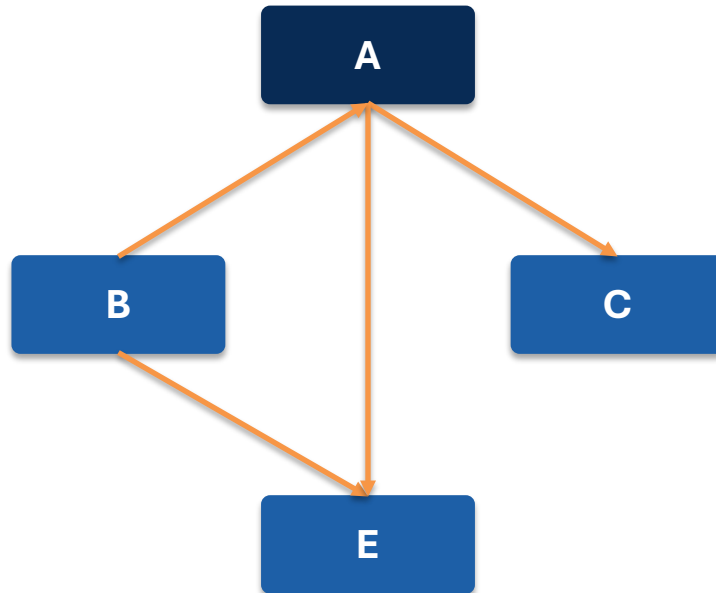
For Delay D
Parents: W and T

Ancestor of D
Rain R

Descendants of R
 W , T , and D

You Try: read the graph

Identify graph relationships before doing any arithmetic.

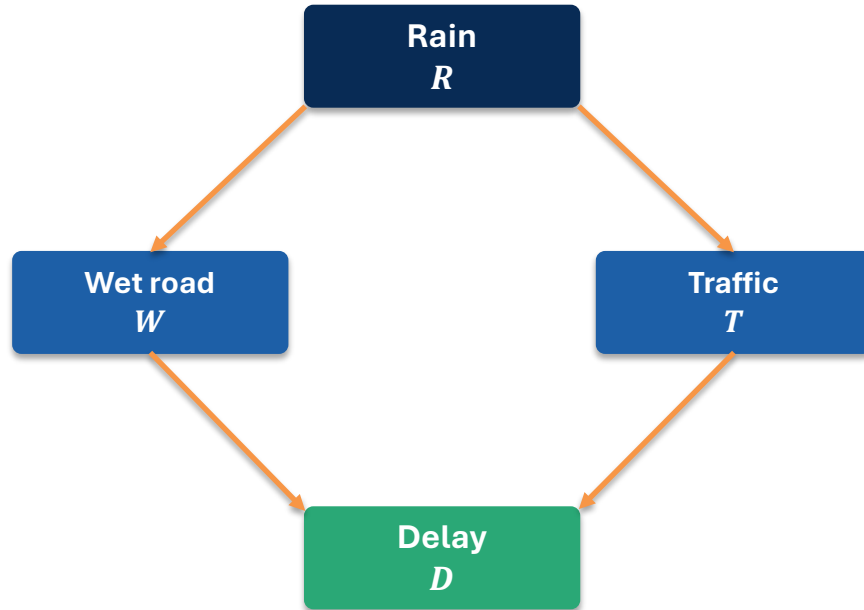


- Parents of E?
- Children of A?
- One ancestor of E?
- One descendant of A?

Answer with graph vocabulary only. No probabilities yet.

Graph structure alone is not enough

The arrow says a relationship exists; it does not give the numerical probabilities.



Question

If it rains, how likely is the road to be wet?

Need

a local probability table

Bayes Net = graph structure + local probability tables.

Root nodes get prior distributions

If a node has no parents, store $P(\text{node})$.

Rain
R

Rain <i>R</i>	$P(R)$
true	.30
false	.70

No parents → prior distribution

One parent → conditional probability table

If W depends directly on R , store $P(W \mid R)$.

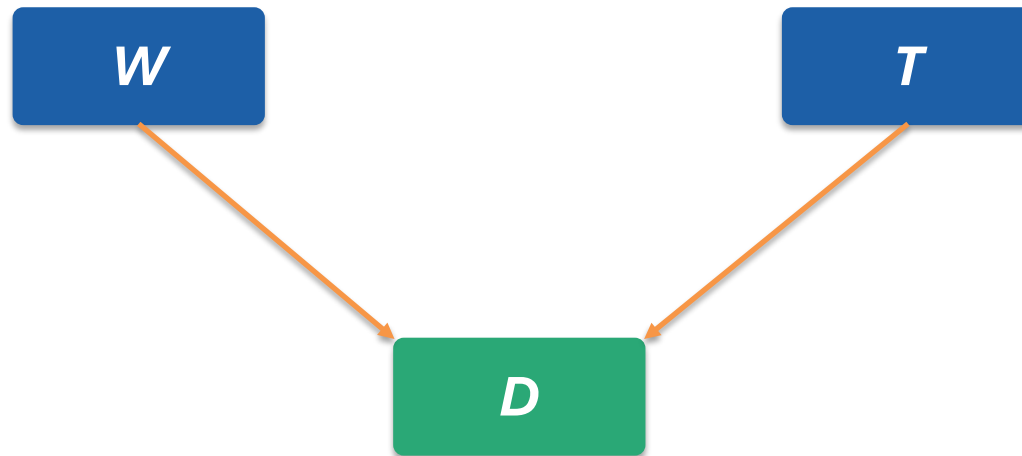


R	$P(W = \text{true} \mid R)$	$P(W = \text{false} \mid R)$
true	.90	.10
false	.15	.85

Each row is a distribution over the child variable for one parent setting.

Two parents → condition on both

Delay depends on wet roads and traffic, so store $P(D \mid W, T)$.

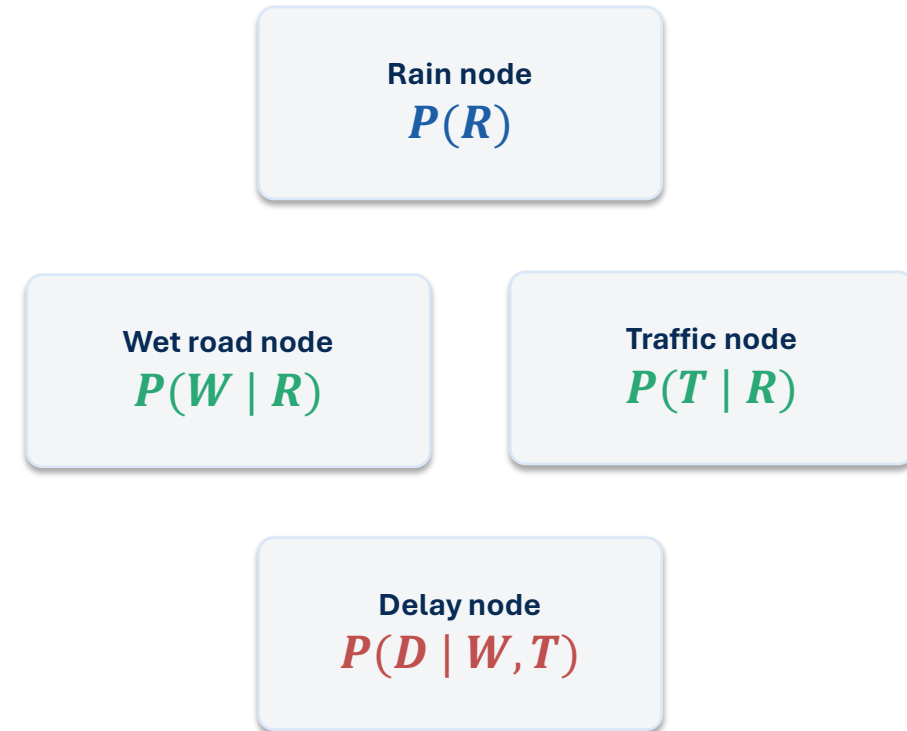
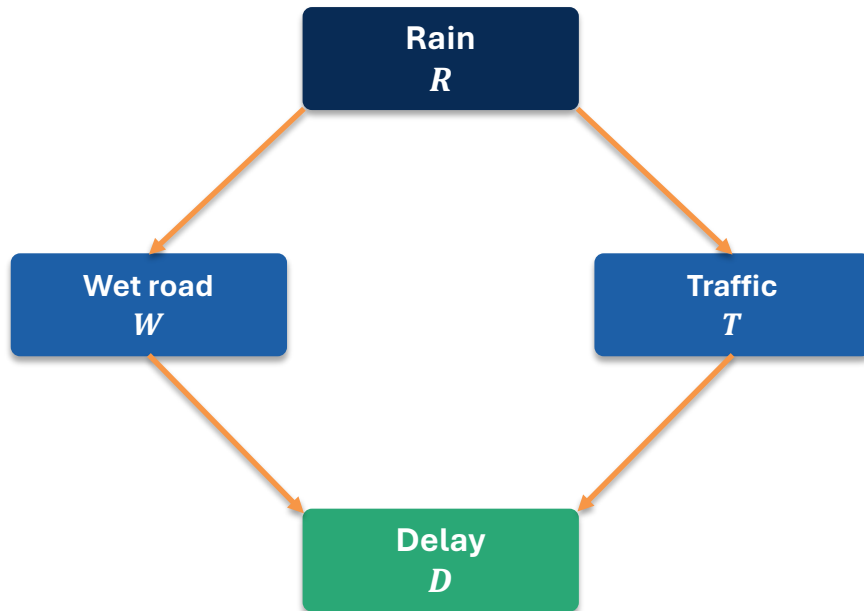


W	T	$P(D=\text{true})$	$P(D=\text{false})$
T	T	.95	.05
T	F	.65	.35
F	T	.70	.30
F	F	.05	.95

Every combination of parent values gets a probability distribution for the child.

Probability tables belong to nodes

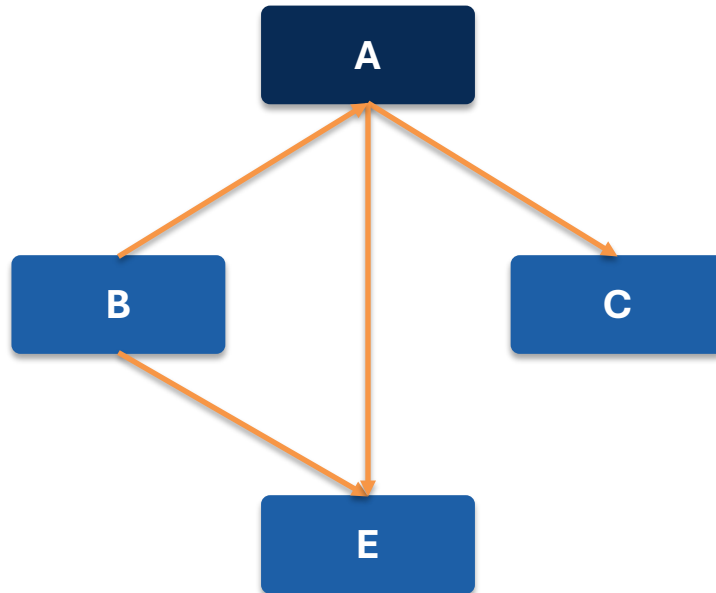
A full Bayesian network includes both the graph and the local tables.



Graph + CPTs = a complete probability model over the variables.

You Try: which CPT belongs here?

Determine which probability tables are needed for each node.



Nodes and Terms

- A:
- B:
- C:
- E:

Rule: each node is conditioned only on its parents.

Primer: recall the product rule

Bayes-net factorization extends an idea we already used.

From L18

$$P(A, B) = P(A)P(B | A)$$

joint = first event × conditional next event

Used in L19

$$P(F, A) = P(A | F)P(F)$$

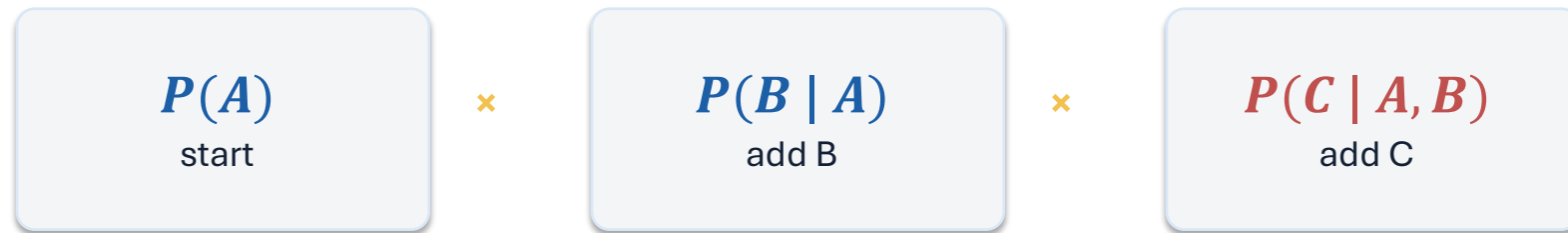
same joint event, written from one direction

Now we extend the same idea from two variables to many variables.

Three variables: the chain rule

Start with one variable, then multiply by each new variable conditioned on what came before.

$$P(A, B, C) = P(A) P(B | A) P(C | A, B)$$



The chain rule is always true; Bayesian networks make it compact by using independence assumptions.

Conditional independence can simplify the chain

The graph can tell us that some earlier variables no longer matter once parents are known.



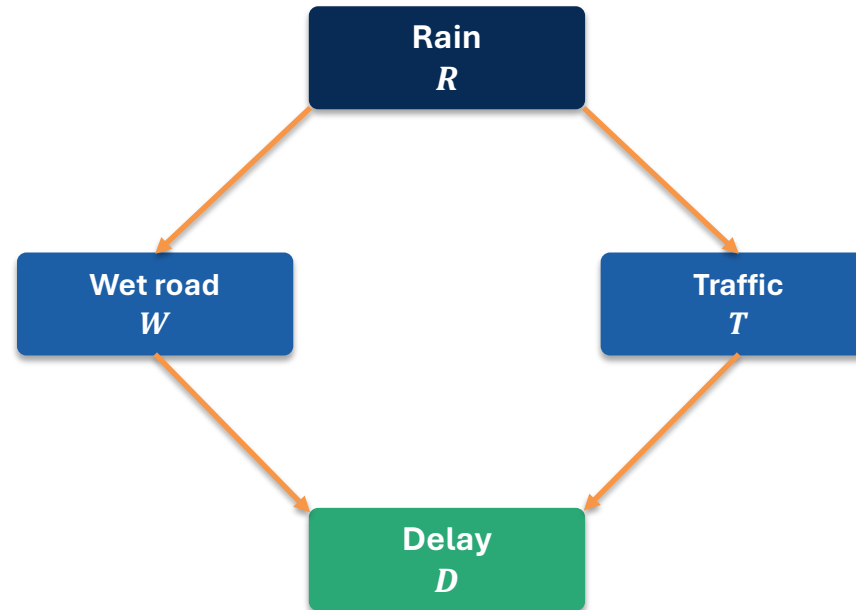
Without simplification
 $P(A)P(B|A)P(C|A, B)$

If $C \perp A | B$
 $P(A)P(B|A)P(C|B)$

Once B is known, A provides no additional information about C in this chain model.

The graph tells us which conditionals to use

For every node, condition only on its parents.



$$P(R, W, T, D) = P(R) P(W | R) P(T | R) P(D | W, T)$$

This is the central meaning of the network.

The full joint is represented as a product of local terms.

General factorization rule

The compact rule is simple once the graph vocabulary is clear.

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \textit{Parents}(X_i))$$

Plain English

Write one probability term per node.

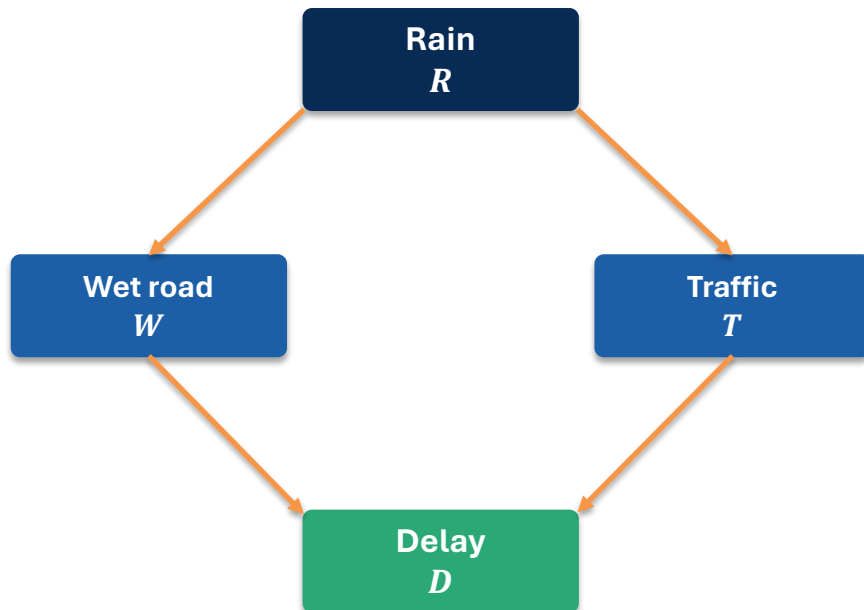
Condition on

only the parents of that node.

Do not condition every node on every earlier variable unless the graph requires it.

A Bayes net assigns probability to full assignments

To score one complete world, multiply the matching local entries.



Assignment
 $r, w, \neg t, d$

$$P(r, w, \neg t, d) = P(r) P(w|r) P(\neg t|r) P(d|w, \neg t)$$

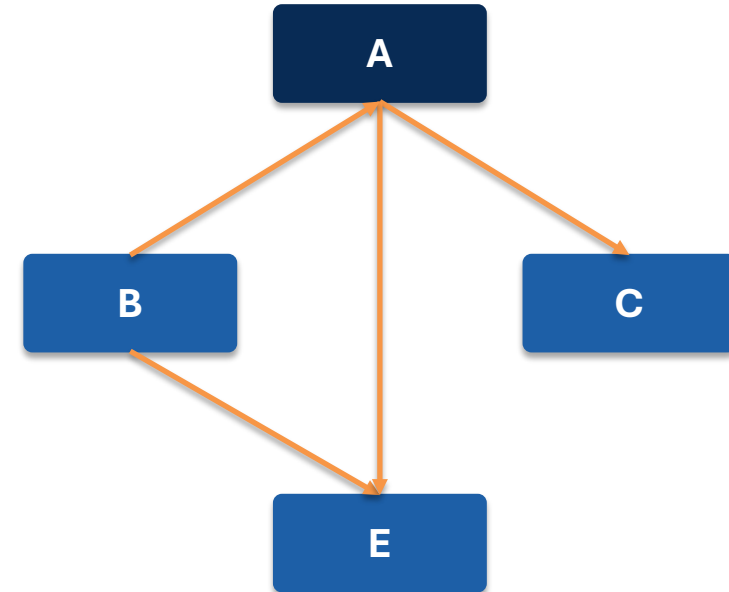
Interpretation

The graph tells us which local probability entries to multiply.

You Try: factor the joint

Use the graph to write one local probability term per node.

Write:
 $P(A,B,C,E) = ?$



Expected structure: $P(B)P(A|B)P(C|A)P(E|A, B)$

Conditional independence: what does it mean?

L18 introduced the phrase. Bayesian networks make it visible.

$$X \perp Y \mid Z$$

Meaning

Once Z is known, learning X provides no additional information about Y .

Warning

This is not the same as saying X and Y can never be related.

Today: learn three basic structures. Full d -separation comes later in advanced AI.

Pattern 1 — Chain

Information flows through the middle variable.

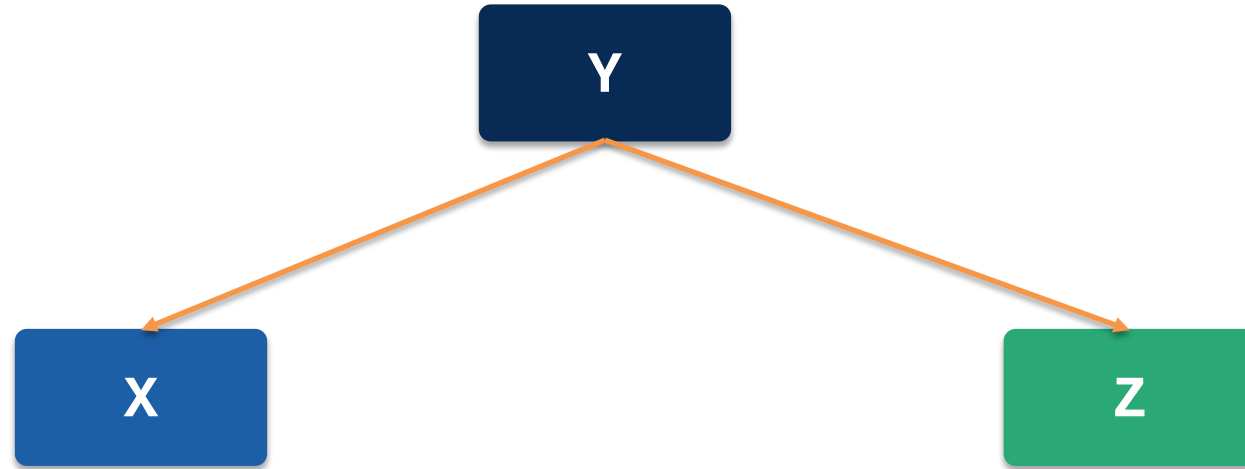


$$X \perp Z \mid Y$$

Once Y is known, X no longer adds information about Z in this chain.

Pattern 2 — Common cause / fork

A shared cause can make two effects appear related.

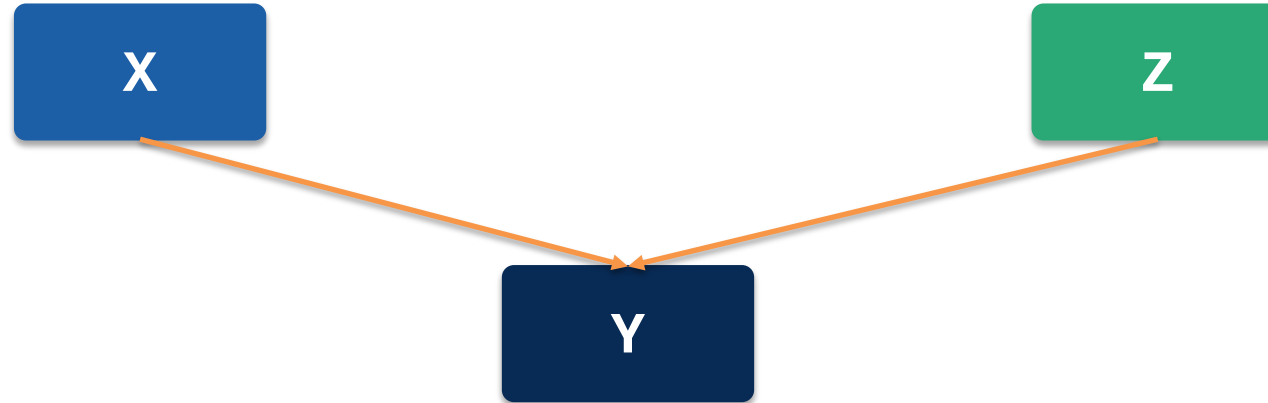


$$X \perp Z \mid Y$$

Once the common cause Y is known, one effect may add no new information about the other.

Pattern 3 — Common effect / collider

This pattern behaves differently from chains and forks.



Explaining away

Observing the common effect can make the possible causes informative about each other.

You Try: which relationship holds?

Use the three motifs to reason about conditional independence.

Chain

$X \rightarrow Y \rightarrow Z$

$X \perp Z \mid Y?$

Fork

$X \leftarrow Y \rightarrow Z$

$X \perp Z \mid Y?$

Collider

$X \rightarrow Y \leftarrow Z$

What changes if Y is observed?

Chains and forks are blocked by observing the middle variable. Colliders are different.

Arrows are not automatically causation

A Bayesian network encodes probabilistic structure; sometimes that structure is causal, but not always.

What arrows do mean
direct probabilistic dependence
and conditional-independence
assumptions

What arrows do not prove
causation by themselves
Modeling choices and domain knowledge matter.

Good causal structure often makes a Bayesian network simpler, but the graph itself is still a model assumption.

Practice: build a Bayesian network

Construct the model from a small scenario.

A smart greenhouse tracks weather, irrigation, soil moisture, and plant stress.
Weather affects soil moisture.
Irrigation affects soil moisture.
Plant stress depends on soil moisture.

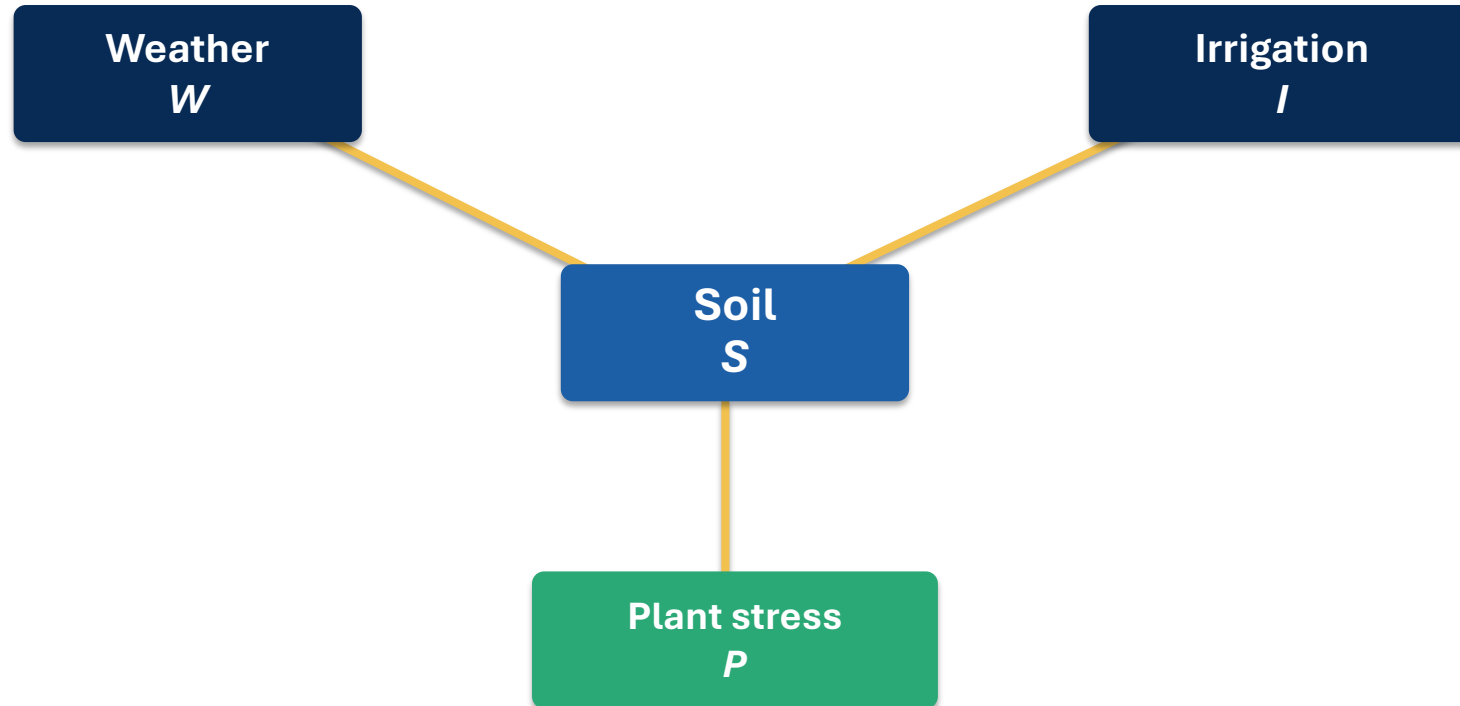
- Draw the network.
- Identify the parents of each node.
- State the CPT each node needs.
- Write the full-joint factorization.

Goal: translate scenario → graph → CPTs → factorization.

Solution: greenhouse network

One reasonable model uses weather and irrigation as causes of soil moisture.

$$\text{Factorization: } P(W, I, S, P) = P(W) P(I) P(S | W, I) P(P | S)$$



This graph assumes plant stress depends on weather and irrigation only through soil moisture.

Bayesian-network toolbox

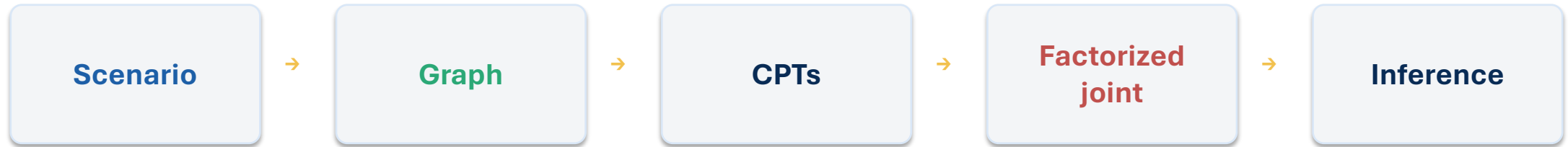
Use this vocabulary to translate between diagrams and probability models.

Representation	Meaning
Node	random variable
Arrow	direct probabilistic dependence
Root node	prior distribution
Non-root node	CPT conditioned on parents
Graph + CPTs	complete Bayesian-network model
Factorization	full joint represented compactly

The graph is useful because it tells us which local probability tables matter.

Synthesis: from scenario to inference

L20 builds the model; L21 uses it to answer probability queries.



Exit ticket

Given $A \rightarrow B$; $B \rightarrow C$; $A, C \rightarrow D$:

Parents of D?

One ancestor of D?

Probability term for D?

Factorize $P(A, B, C, D)$.

Next: once the network exists, how do we compute answers from it?